

A conditional diffusion emulator of Veros ACC equilibrium states: first end-to-end results

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Summary

We trained a conditional denoising diffusion model to generate the post-spin-up temperature and salinity state of the Veros ACC channel configuration directly from the two coefficients of its eddy-kinetic-energy closure, c_k and c_ϵ . On ten held-out parameter pairs the eight-sample ensemble mean has an RMSE of 0.030 K against the true 20-year mean state, with a domain-mean bias below 1 mK, and the generated response over the full 10×10 parameter grid reproduces the true one. The choice of normalisation decides whether conditioning works at all: DINO-Fusion’s per-level scaling makes the model collapse to the mean state (0.154 K), per-cell anomaly scaling does not. The emulator beats the nearest training run (0.037 K) but not yet the average of the four neighbouring runs on the grid (0.020 K). The full chain costs under one A100 hour.

1 Data

One hundred Veros ACC runs on a complete 10×10 log-spaced grid, $c_k \in [0.0125, 0.8]$ and $c_\epsilon \in [0.0875, 5.6]$, 100 years each with 30-day output on a $30 \times 42 \times 15$ interior grid (zonal, meridional, vertical). We keep temperature and salinity for the last 20 years of every run (241 snapshots per run). Three properties of the ensemble shape the problem:

- Salinity is exactly 35 psu in every water cell of every run; it is carried through the pipeline as a second field but contains no information.
- The end states are nearly deterministic: after removing the residual drift, the grid-point temporal standard deviation is 0.002 K, against 0.05 K between neighbouring runs on the grid and up to 1.5 K between extremes. The runs still drift by 0.03 to 0.12 K in domain-mean temperature over the last 20 years, so the target is the state at year 100 rather than a true equilibrium.
- The two parameters act almost as one: $\log c_k - 1.05 \log c_\epsilon$ explains 99 % of the variance of the domain-mean temperature, and the low- c_k/c_ϵ part of the grid saturates to one state. The response lives between 600 and 800 m and near the bottom, not at the restored surface.

Ten interior grid points were held out at random before any statistic was computed; the remaining 90 runs (21 690 snapshots) form the training set.

2 Method

The pipeline is a fork of the DINO-Fusion diffusion code adapted to a parameter ensemble. Temperature and salinity are stacked into 30 channels, normalised, set to zero on the fixed land mask, and zero-padded from 42×30 to 48×32 so the UNet can halve the grid four times. Two normalisations

were trained with otherwise identical settings:

$$\begin{aligned} \text{per level (DINO): } x' &= (x - \mu_z) / (3\sigma_z), \\ \text{per cell (anomaly): } x' &= (x - \mu_{zyx}) / (3 \max(\sigma_{zyx}, 0.05)), \end{aligned}$$

with statistics from the training runs only. The floor keeps the constant salinity at exactly zero. The denoiser is a four-level UNet2DModel (64, 64, 128, 128 channels, 6.9 M parameters). The standardised $(\log c_k, \log c_\epsilon)$ pair passes through a small MLP whose output is added to the timestep embedding. Training is the standard DDPM ϵ -prediction objective with 1000 steps, a squared-cosine schedule, sample clipping, AdamW at 10^{-4} with cosine decay, batch 32, 20 000 steps and an EMA of the weights. Sampling runs the 1000-step DDPM loop with the EMA weights and re-imposes zeros on land and padding after every step.

3 Evaluation

For each held-out run we draw eight samples at its (c_k, c_ϵ) and compare, on water cells, with the true time-mean state of the last 20 years. Baselines: the mean training state, the nearest training run in standardised log-parameter space, and the average of the axis-neighbour training runs on the grid. We also generate four samples for every one of the 100 grid points to compare the map of domain-mean temperature with the truth.

	anomaly	per level
diffusion, ensemble mean of 8: RMSE (K)	0.030 ± 0.009	0.154 ± 0.018
diffusion, single sample: RMSE (K)	0.072	0.188
neighbour average: RMSE (K)	0.020 ± 0.011	0.020 ± 0.011
nearest training run: RMSE (K)	0.037 ± 0.027	0.037 ± 0.027
mean training state: RMSE (K)	0.155 ± 0.030	0.155 ± 0.030
diffusion: domain-mean bias (K)	-0.0002	-0.028
ensemble spread / true within-window spread (K)	0.049 / 0.031	0.105 / 0.031
interfaces with T decreasing upward, generated / truth	8.4 % / 7.7 %	9.5 % / 7.7 %
max salinity error (psu)	0.028	0.010
domain-mean T map, RMSE on hold-out / all runs (K)	0.012 / 0.045	0.030 / 0.049

Table 1: Hold-out metrics, mean $\pm$ standard deviation over the ten held-out runs, water cells only. Baselines are identical in both columns because they do not depend on the model.

4 Findings

1. **Normalisation decides whether conditioning works.** The run-to-run signal is about 2 % of the spatial structure in the upper ocean, so under per-level scaling the conditional model sees almost no signal and reproduces the training mean; its parameter map is flat and noisy. Per-cell anomaly scaling makes the signal order one everywhere and the model follows the parameters, including the warm high- c_k , low- c_ϵ corner (Figure 1).
2. **Better than the nearest run, not yet better than grid interpolation.** The ensemble mean reaches 0.030 K against 0.020 K for the neighbour average. Single samples are 1.6 times more dispersed than the true within-window variability.
3. **The error sits below 1200 m** (Figure 2), as a warm bias of one to two tenths of a kelvin in the deepest layers, where the anomaly distribution has the heaviest tails and about 2 % of training values fall outside the clipping range. The cold bias at the extreme corner run is the regime change at the edge of the training set.
4. **Plumbing checks pass.** Salinity returns within 0.03 psu of 35 without any constraint, land is exact, and the fraction of temperature inversions matches the truth. Figure 3 shows random samples next to the true state at three depths.

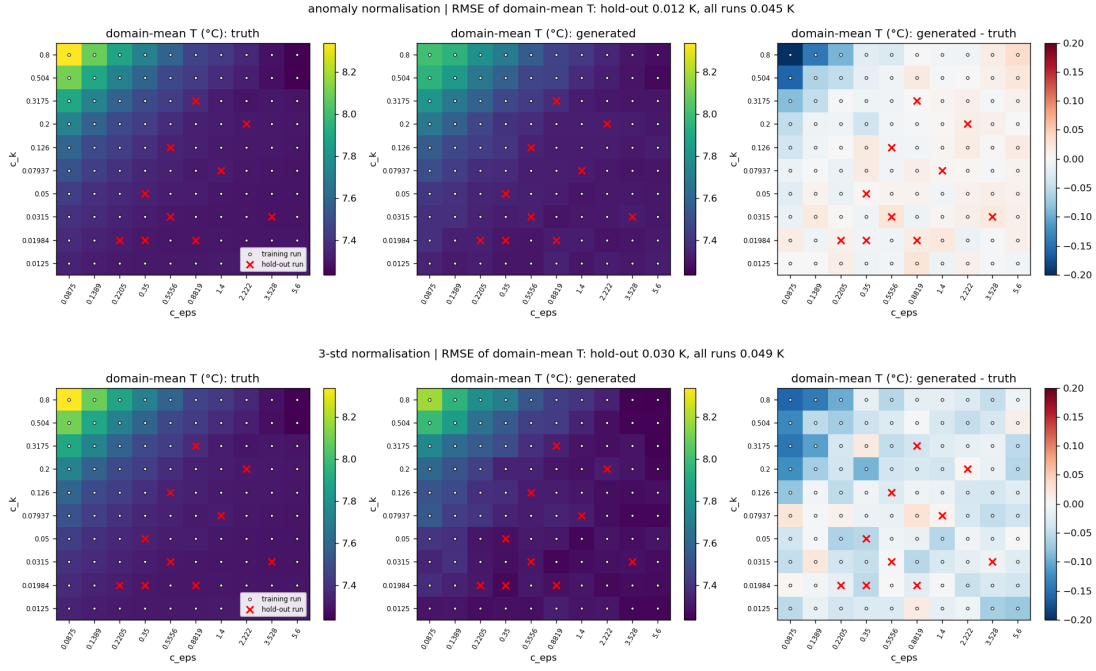


Figure 1: Domain-mean temperature over the parameter grid: truth, generated, and difference. Top: anomaly normalisation. Bottom: per-level normalisation. White dots mark training runs, red crosses the held-out runs.

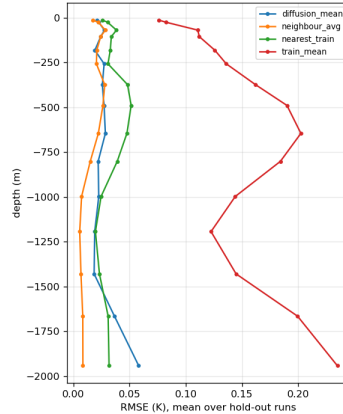


Figure 2: Anomaly normalisation: RMSE against the truth by depth, mean over the held-out runs, for the diffusion ensemble mean and the three baselines.

5 Next steps

Raise the anomaly scale to four standard deviations or disable clipping; draw 32 samples per condition, since the ensemble mean is what competes with the baselines; hold out the whole $c_k = 0.8$ row to test the regime corner; try classifier-free guidance to sharpen a small conditioning signal; and add the year as a third condition to absorb the residual drift. The longer-term target is the same pipeline on a global Veros configuration, where active salinity, seasonal forcing and multi-century spin-up make the generative formulation and the density constraints of DINO-Fusion relevant.

Reproducibility. Branch `miniveros` of the DINO-Fusion repository, package `miniveros/`, commit `f912e66` for the runs reported here. Extraction takes 2.5 minutes on 8 CPU cores, each training 22 minutes on one A100, generation and evaluation 3.5 minutes per run; the whole dependency chain ran on the development queue in under one GPU hour.

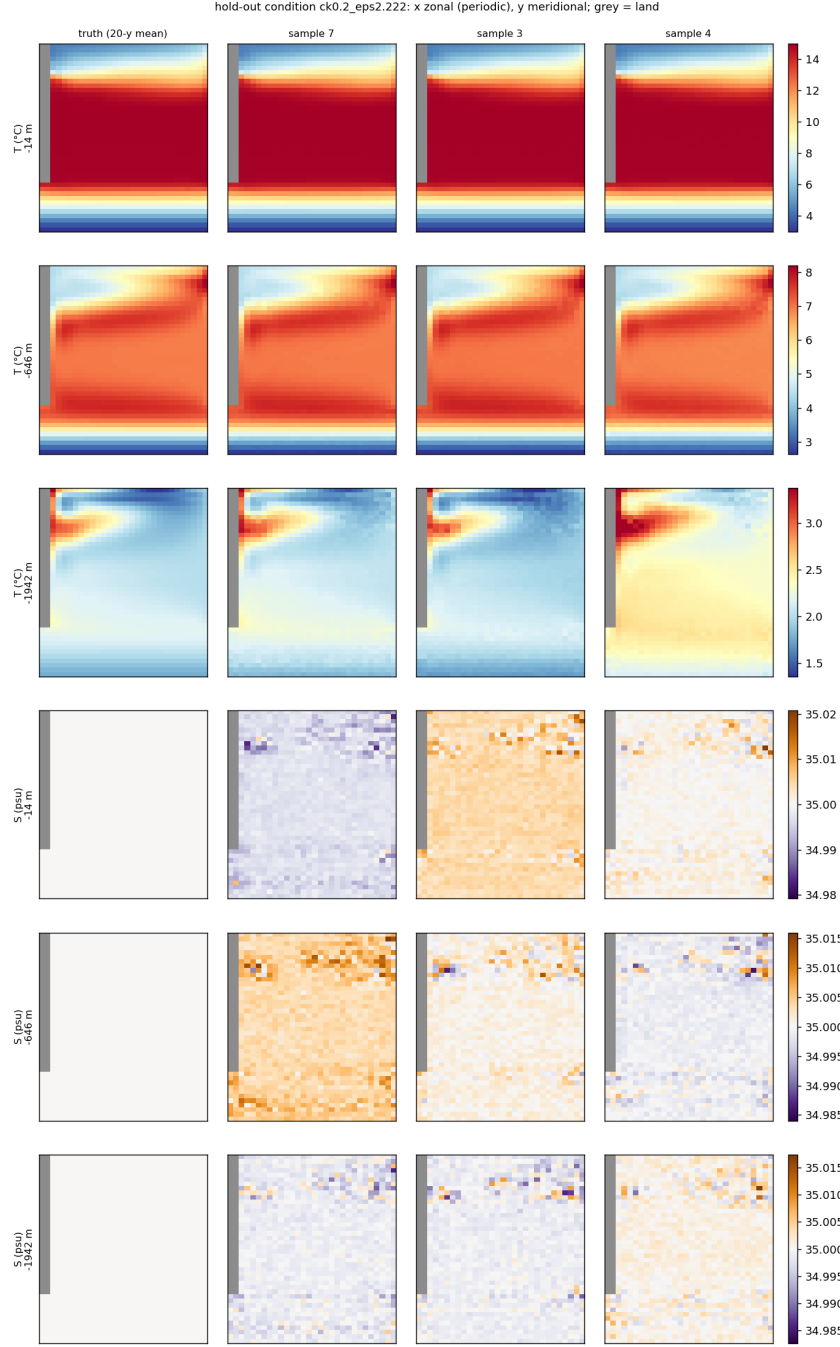


Figure 3: Visual check for one held-out condition ($c_k = 0.2$, $c_\epsilon = 2.222$, anomaly model): the true 20-year mean state (left column) and three random samples at the surface, at about 650 m and at the bottom, for temperature (top three rows) and salinity (bottom three rows). Surface and mid-depth samples are indistinguishable from the truth; one sample departs visibly at the bottom, where the RMSE and the spread are largest. Salinity is 35 psu everywhere in the truth; the salinity panels show the residual noise of the generator on a colour scale of a few hundredths of a psu. Grey is land.